

SOC ANALYST PORTFOLIO CASE STUDY

Suricata Network IDS and Wazuh SIEM Integration

Deploying a network sensor, validating default and custom detections, and preserving end-to-end SIEM evidence

VALIDATED OUTCOME

Twenty controlled ICMP requests and one custom HTTP request produced 21 validated end-to-end alerts.

Analyst	Ryan Bryant
Exercise date	August 28, 2026
Environment	Authorized VirtualBox SOC home lab
Classification	Portfolio demonstration — no production systems

AUTHORIZED-LAB SAFETY NOTE

All traffic generation, sensor tuning, SIEM configuration, and evidence collection occurred inside a controlled private network owned and operated by the analyst.

Executive Summary

The analyst deployed Suricata 7.0.3 on LINUX-WEBSERVER01, corrected the AF_PACKET sensor interface to enp0s3, narrowed HOME_NET to the protected host, and configured the Wazuh agent to ingest EVE JSON. A 20-request ICMP exercise triggered 20 default Suricata alerts that Wazuh indexed with the expected source, destination, signature, agent, and rule metadata. The analyst then engineered SID 1000001 for a lab-only HTTP admin-path request; one controlled request generated one Suricata alert and one correlated Wazuh event with method, URI, status, and network context. Fifteen collected artifacts passed SHA-256 verification.

DISPOSITION

End-to-end network detection and SIEM telemetry validated. No blocking, production compromise, or internet-scale performance claim was made.

Objective and Scope

- Primary objective: Prove that Suricata network alerts travel from the protected endpoint into Wazuh with usable context.
- Detection scope: Validate one default ICMP signature and one analyst-authored HTTP URI signature.
- Collection scope: EVE JSON produced on 10.10.10.4 and ingested by the Wazuh agent.
- Safety scope: Authorized private-lab traffic only; no exploitation, persistence, malware, or traffic blocking.

Lab Environment

Role	System	Address / Function
Test source	Parrot Security	10.10.10.5 — controlled ICMP and HTTP requests
Protected endpoint	LINUX-WEBSERVER01	10.10.10.4 — Apache, Suricata sensor, Wazuh agent
SIEM	Wazuh server	10.10.10.3 — manager; dashboard via 192.168.56.101
Network	VirtualBox NAT Network	10.10.10.0/24 — isolated lab segment

Technical Workflow and Sensor Configuration

01 GENERATE Parrot sends approved test traffic	02 DETECT Suricata inspects enp0s3 and writes EVE JSON	03 CORRELATE Wazuh decodes, indexes, and exposes fields
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The sensor ran in IDS mode. The protected endpoint remained available throughout the exercise; Suricata observed traffic and generated alerts but did not block connections.

Key Configuration

Control	Validated setting	Reason
Capture	AF_PACKET on enp0s3	Matches the Ubuntu interface carrying 10.10.10.4/24
Scope	HOME_NET = [10.10.10.4/32]	Treats only the protected web server as the monitored asset
External	EXTERNAL_NET = !\$HOME_NET	Classifies the Parrot source as outside the protected host
Rules	suricata.rules + local.rules	Combines community coverage with analyst-authored logic
Output	/var/log/suricata/eve.json	Provides structured JSON for downstream collection
SIEM collector	log_format json	Preserves Suricata fields during Wazuh ingestion

```
Suricata and Wazuh Integration Status
Timestamp: 2026-08-28T20:52:26+00:00
Hostname: linux-webserver01

This is Suricata version 7.0.3 RELEASE
Suricata service: active
Suricata startup setting: enabled
Wazuh agent: active

Sensor interface:
enp0s3          UP          10.10.10.4/24 metric 100 fe80::a00:27ff:fe91:c92a/64

Network variables:
18:  HOME_NET: "[10.10.10.4/32]"
24:  EXTERNAL_NET: "!$HOME_NET"

Loaded rule files:
2152:rule-files:
2153-  - suricata.rules
2154-  - local.rules
2155-

Custom rule:
alert http $EXTERNAL_NET any -> $HOME_NET 80 (msg:"SOC LAB Suspicious Admin Path Request"; flow:established,to_server; http.method; content:"GET"; http.uri; content:"/soc-lab-admin-test"; nocase; classtype:attempted-recon; sid:1000001; rev:1;)

Suricata event log:
-rw-r--r-- 1 root root 2.7M Aug 28 20:52 /var/log/suricata/eve.json

Wazuh Suricata collector:
<localfile>
  <log_format>json</log_format>
  <location>/var/log/suricata/eve.json</location>
</localfile>
</ossec_config>
socstudent@linux-webserver01:~$ |
```

Figure 1. Consolidated status evidence shows Suricata 7.0.3 active, enp0s3 monitored, the custom rule loaded, and the Wazuh JSON collector configured.

Baseline and Default Detection Validation

Before generating detection traffic, the analyst verified basic service health from Parrot: three ICMP replies were returned with 0% loss and Apache returned HTTP 200. This established that the protected endpoint and web service were reachable before alert testing.

```
[user@parrot]~$
$
echo "Timestamp: $(date -Is)"
echo "Source: 10.10.10.5"
echo "Target: 10.10.10.4"
ping -c 3 10.10.10.4
curl -s -o /dev/null -w "HTTP status: %{http_code}\n" http://10.10.10.4/
} | tee "$HOME/SOC-Labs/Suricata-Wazuh-IDS/recon/01-benign-connectivity-baseline.txt"
Timestamp: 2026-08-28T20:19:22+00:00
Source: 10.10.10.5
Target: 10.10.10.4
PING 10.10.10.4 (10.10.10.4) 56(84) bytes of data.
64 bytes from 10.10.10.4: icmp_seq=1 ttl=64 time=1.71 ms
64 bytes from 10.10.10.4: icmp_seq=2 ttl=64 time=0.989 ms
64 bytes from 10.10.10.4: icmp_seq=3 ttl=64 time=0.732 ms

--- 10.10.10.4 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2003ms
rtt min/avg/max/mdev = 0.732/1.145/1.714/0.415 ms
HTTP status: 200
[user@parrot]~$
$
```

Figure 2. Benign connectivity baseline: 3/3 ICMP replies and HTTP 200 from the protected endpoint.

Controlled ICMP Test

20 REQUESTS ICMP echo requests sent from 10.10.10.5	20 SURICATA ALERTS Default SID 2100366 matched every request	0% PACKET LOSS The sensor detected without disrupting availability
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The default signature GPL ICMP PING *NIX (SID 2100366) categorized the activity as Misc activity with Suricata severity 3. Each event preserved source 10.10.10.5, destination 10.10.10.4, protocol ICMP, and the event timestamp.

```
Recent Suricata alerts involving Parrot:
{"timestamp":"2026-08-28T20:32:19.982852+0000","src_ip":"10.10.10.5","dest_ip":"10.10.10.4","proto":"ICMP","signature_id":2100366,"signature":"GPL ICMP PING *NIX","category":"Misc activity","severity":3}
{"timestamp":"2026-08-28T20:32:20.983878+0000","src_ip":"10.10.10.5","dest_ip":"10.10.10.4","proto":"ICMP","signature_id":2100366,"signature":"GPL ICMP PING *NIX","category":"Misc activity","severity":3}
{"timestamp":"2026-08-28T20:32:22.020074+0000","src_ip":"10.10.10.5","dest_ip":"10.10.10.4","proto":"ICMP","signature_id":2100366,"signature":"GPL ICMP PING *NIX","category":"Misc activity","severity":3}
{"timestamp":"2026-08-28T20:32:23.044763+0000","src_ip":"10.10.10.5","dest_ip":"10.10.10.4","proto":"ICMP","signature_id":2100366,"signature":"GPL ICMP PING *NIX","category":"Misc activity","severity":3}
{"timestamp":"2026-08-28T20:32:24.045796+0000","src_ip":"10.10.10.5","dest_ip":"10.10.10.4","proto":"ICMP","signature_id":2100366,"signature":"GPL ICMP PING *NIX","category":"Misc activity","severity":3}
{"timestamp":"2026-08-28T20:32:25.046797+0000","src_ip":"10.10.10.5","dest_ip":"10.10.10.4","proto":"ICMP","signature_id":2100366,"signature":"GPL ICMP PING *NIX","category":"Misc activity","severity":3}
{"timestamp":"2026-08-28T20:32:26.049927+0000","src_ip":"10.10.10.5","dest_ip":"10.10.10.4","proto":"ICMP","signature_id":2100366,"signature":"GPL ICMP PING *NIX","category":"Misc activity","severity":3}
{"timestamp":"2026-08-28T20:32:27.050115+0000","src_ip":"10.10.10.5","dest_ip":"10.10.10.4","proto":"ICMP","signature_id":2100366,"signature":"GPL ICMP PING *NIX","category":"Misc activity","severity":3}
{"timestamp":"2026-08-28T20:32:28.099762+0000","src_ip":"10.10.10.5","dest_ip":"10.10.10.4","proto":"ICMP","signature_id":2100366,"signature":"GPL ICMP PING *NIX","category":"Misc activity","severity":3}
{"timestamp":"2026-08-28T20:32:29.102772+0000","src_ip":"10.10.10.5","dest_ip":"10.10.10.4","proto":"ICMP","signature_id":2100366,"signature":"GPL ICMP PING *NIX","category":"Misc activity","severity":3}
{"timestamp":"2026-08-28T20:32:30.147922+0000","src_ip":"10.10.10.5","dest_ip":"10.10.10.4","proto":"ICMP","signature_id":2100366,"signature":"GPL ICMP PING *NIX","category":"Misc activity","severity":3}
{"timestamp":"2026-08-28T20:32:31.149037+0000","src_ip":"10.10.10.5","dest_ip":"10.10.10.4","proto":"ICMP","signature_id":2100366,"signature":"GPL ICMP PING *NIX","category":"Misc activity","severity":3}
{"timestamp":"2026-08-28T20:32:32.150711+0000","src_ip":"10.10.10.5","dest_ip":"10.10.10.4","proto":"ICMP","signature_id":2100366,"signature":"GPL ICMP PING *NIX","category":"Misc activity","severity":3}
{"timestamp":"2026-08-28T20:32:33.152760+0000","src_ip":"10.10.10.5","dest_ip":"10.10.10.4","proto":"ICMP","signature_id":2100366,"signature":"GPL ICMP PING *NIX","category":"Misc activity","severity":3}
{"timestamp":"2026-08-28T20:32:34.154794+0000","src_ip":"10.10.10.5","dest_ip":"10.10.10.4","proto":"ICMP","signature_id":2100366,"signature":"GPL ICMP PING *NIX","category":"Misc activity","severity":3}
{"timestamp":"2026-08-28T20:32:35.156518+0000","src_ip":"10.10.10.5","dest_ip":"10.10.10.4","proto":"ICMP","signature_id":2100366,"signature":"GPL ICMP PING *NIX","category":"Misc activity","severity":3}
{"timestamp":"2026-08-28T20:32:36.156975+0000","src_ip":"10.10.10.5","dest_ip":"10.10.10.4","proto":"ICMP","signature_id":2100366,"signature":"GPL ICMP PING *NIX","category":"Misc activity","severity":3}
{"timestamp":"2026-08-28T20:32:37.159593+0000","src_ip":"10.10.10.5","dest_ip":"10.10.10.4","proto":"ICMP","signature_id":2100366,"signature":"GPL ICMP PING *NIX","category":"Misc activity","severity":3}
{"timestamp":"2026-08-28T20:32:38.161732+0000","src_ip":"10.10.10.5","dest_ip":"10.10.10.4","proto":"ICMP","signature_id":2100366,"signature":"GPL ICMP PING *NIX","category":"Misc activity","severity":3}
{"timestamp":"2026-08-28T20:32:39.162990+0000","src_ip":"10.10.10.5","dest_ip":"10.10.10.4","proto":"ICMP","signature_id":2100366,"signature":"GPL ICMP PING *NIX","category":"Misc activity","severity":3}
socstudent@linux-webserver01:~$
```

Figure 3. EVE JSON contained the expected 20 alert records for SID 2100366.

Wazuh Correlation and Analyst Visibility

Wazuh indexed all 20 ICMP events under built-in rule 86601 at level 3 in the ids and suricata groups. The dashboard exposed the protected agent, Suricata signature metadata, source and destination addresses, interface, protocol, and firing count. The Wazuh rule level reflects SIEM prioritization and is separate from Suricata's event severity.

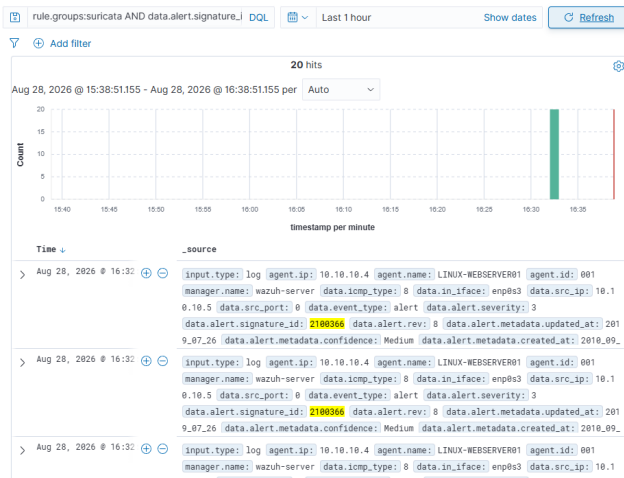


Figure 4. Discover returned exactly 20 hits for SID 2100366.

data.icmp_code	0
data.icmp_type	8
data.in_iface	enp0s3
data.pkt_src	wire/pcap
data.proto	ICMP
data.src_ip	10.10.10.5
data.src_port	0
data.timestamp	Aug 28, 2026 @ 16:32:38.161
decoder.name	json
id	1787949159.338313
input.type	log
location	/var/log/suricata/eve.json
manager.name	wazuh-server
rule.description	Suricata: Alert - GPL ICMP PING *NIX
rule.firedtimes	19
rule.groups	ids, suricata
rule.id	86601
rule.level	3
rule.mail	false
timestamp	Aug 28, 2026 @ 16:32:39.793

Figure 5. Expanded evidence confirmed rule 86601, level 3, source, agent, and groups.

ANALYST CONCLUSION

The default network signature traveled from packet capture to structured endpoint telemetry and into a searchable SIEM alert without losing core context.

Custom Detection Engineering

To demonstrate analyst-authored detection logic, the analyst created a local HTTP rule for the exact case-insensitive URI /soc-lab-admin-test. The rule required an established client-to-server flow, the GET method, and traffic entering TCP 80 from EXTERNAL_NET to the protected HOME_NET host.

```
alert http $EXTERNAL_NET any -> $HOME_NET 80 (msg:"SOC LAB Suspicious Admin Path Request"; flow:established,to_server;
http.method; content:"GET"; http.uri; content:"/soc-lab-admin-test"; nocase; classtype:attempted-recon; sid:1000001; rev:1;)
```

The rule was stored in /var/lib/suricata/rules/local.rules, added to rule-files, syntax-tested, and loaded only after validation succeeded.

```
socstudent@linux-webserver01:~$ sudo cp --preserve=all \
/etc/suricata/suricata.yaml \
/etc/suricata/suricata.yaml.pre-local-rule

printf '%s\n' \
'alert http $EXTERNAL_NET any -> $HOME_NET 80 (msg:"SOC LAB Suspicious Admin Path Request"; flow:established,to_server;
http.method; content:"GET"; http.uri; content:"/soc-lab-admin-test"; nocase; classtype:attempted-recon; sid:1000001; rev
:1;)' |
sudo tee /var/lib/suricata/rules/local.rules >/dev/null

sudo chmod 644 /var/lib/suricata/rules/local.rules

sudo grep -qE '^[[[:space:]]*~[[[:space:]]*]+local.rules$' \
/etc/suricata/suricata.yaml ||
sudo sed -i '/^[[[:space:]]*~ suricata.rules$/a \ - local.rules' \
/etc/suricata/suricata.yaml

echo "Loaded rule files:"
sudo grep -nA3 'rule-files:' /etc/suricata/suricata.yaml

echo "Custom rule:"
sudo cat /var/lib/suricata/rules/local.rules

sudo suricata -T -c /etc/suricata/suricata.yaml
Loaded rule files:
2152:rule-files:
2153:  - suricata.rules
2154:  - local.rules
2155:
Custom rule:
alert http $EXTERNAL_NET any -> $HOME_NET 80 (msg:"SOC LAB Suspicious Admin Path Request"; flow:established,to_server; h
ttp.method; content:"GET"; http.uri; content:"/soc-lab-admin-test"; nocase; classtype:attempted-recon; sid:1000001; rev:
1;)
i: suricata: This is Suricata version 7.0.3 RELEASE running in SYSTEM mode
i: suricata: Configuration provided was successfully loaded. Exiting.
socstudent@linux-webserver01:~$
```

Figure 6. Suricata loaded local.rules and accepted the custom rule during configuration validation.

Controlled Result

Parrot requested http://10.10.10.4/soc-lab-admin-test. Apache returned 404 because the path intentionally did not exist; the IDS still generated exactly one alert because the request URI itself matched the detection logic.

```
socstudent@linux-webserver01:~$ sudo jq -c '
select(
  .event_type == "alert" and
  .alert.signature_id == 1000001 and
  .src_ip == "10.10.10.5" and
  .dest_ip == "10.10.10.4"
)
' /var/log/suricata/eve.json |
tee "$HOME/suricata-custom-rule-alert.jsonl" >/dev/null

wc -l "$HOME/suricata-custom-rule-alert.jsonl"

jq -c '{
timestamp,
src_ip,
dest_ip,
dest_port,
app_proto,
signature_id: .alert.signature_id,
signature: .alert.signature,
category: .alert.category,
severity: .alert.severity,
url: .http.url,
status: .http.status
}' "$HOME/suricata-custom-rule-alert.jsonl"
1 /home/socstudent/suricata-custom-rule-alert.jsonl
{"timestamp":"2026-08-28T20:47:55.100151+0000","src_ip":"10.10.10.5","dest_ip":"10.10.10.4","dest_port":80,"app_proto":"
http","signature_id":1000001,"signature":"SOC LAB Suspicious Admin Path Request","category":"Attempted Information Leak"
,"severity":2,"url":"/soc-lab-admin-test","status":404}
socstudent@linux-webserver01:~$
```

Figure 7. The raw Suricata alert preserved SID 1000001, URI, HTTP status, source, destination, category, and severity.

Custom Alert Validation in Wazuh

1 CONTROLLED REQUEST GET /soc-lab-admin-test	1 SURICATA ALERT SID 1000001, severity 2	1 WAZUH EVENT Rule 86601, level 3
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The Wazuh event retained the custom signature name and identifier along with HTTP method, URI, user agent, status 404, destination port 80, source 10.10.10.5, and destination 10.10.10.4. This made the alert directly usable for triage without returning to the raw packet stream.

Aug 28, 2026 @ 16:47

input.type: log agent.ip: 10.10.10.4 agent.name: LINUX-WEBSERVER01 agent.id: 001 manager.name: wazuh-server data.tx_id: 0 data.app.proto: http data.in_iface: enp0s3 data.src_ip: 10.10.10.5 data.src_port: 49404 data.event.type: alert data.alert.severity: 2 data.alert.signature_id: 1000001 data.alert.rev: 1 data.alert.gid: 1 data.alert.signature: SOC LAB Suspicious Admin Path Request

Expanded document

Table JSON

_index	wazuh-alerts-4.x-2026.08.28
agent.id	001
agent.ip	10.10.10.4
agent.name	LINUX-WEBSERVER01
data.alert.action	allowed
data.alert.category	Attempted Information Leak
data.alert.gid	1
data.alert.rev	1
data.alert.severity	2
data.alert.signature	SOC LAB Suspicious Admin Path Request
data.alert.signature_id	1000001
data.app.proto	http

Figure 8. Wazuh identified the custom signature, severity, agent, and direction.

data.http.status	404
data.http.url	/soc-lab-admin-test
data.in_iface	enp0s3
data.pkt_src	wire/pcap
data.proto	TCP
data.src_ip	10.10.10.5
data.src_port	49404
data.timestamp	Aug 28, 2026 @ 16:47:55.100
data.tx_id	0
decoder.name	json
id	1787950076.386693
input.type	log
location	/var/log/suricata/eve.json
manager.name	wazuh-server
rule.description	Suricata: Alert - SOC LAB Suspicious Admin Path Request
rule.firedtimes	21
rule.groups	ids, Suricata
rule.id	86601
rule.level	3
rule.mail	false

Figure 9. URL, status, source, destination, and rule metadata remained searchable.

Evidence Integrity

The final package preserved 10 evidence files, 3 reconnaissance/test records, and 2 timestamps. A SHA-256 manifest recorded all 15 artifacts; verification returned OK for every file. The manifest was stored separately and was not recursively included in its own hash list.

```
[user@parrot]--[~/SOC-Labs/Suricata-Wazuh-IDS]
$sha256sum -c evidence/SHA256SUMS.txt
./evidence/suricata-custom-rule-alert.jsonl: OK
./evidence/suricata-defaults.conf: OK
./evidence/suricata-icmp-alerts.jsonl: OK
./evidence/suricata-local.rules: OK
./evidence/suricata-preinstall-baseline.txt: OK
./evidence/suricata-service-config-status.txt: OK
./evidence/suricata-wazuh-custom-alert.jsonl: OK
./evidence/suricata-wazuh-icmp-alerts.jsonl: OK
./evidence/suricata-wazuh-preintegration-baseline.txt: OK
./evidence/suricata.yaml: OK
./notes/lab-end.txt: OK
./notes/lab-start.txt: OK
./recon/01-benign-connectivity-baseline.txt: OK
./recon/02-controlled-icmp-test.txt: OK
./recon/03-custom-http-rule-test.txt: OK
[user@parrot]--[~/SOC-Labs/Suricata-Wazuh-IDS]
$
```

Figure 10. All 15 Suricata–Wazuh lab artifacts passed SHA-256 verification.

Evidence-to-Conclusion Matrix

Evidence	Observed fact	Analyst conclusion
Baseline record	3/3 ping replies and HTTP 200	Protected service was healthy before detection testing
Suricata ICMP export	20 SID 2100366 alerts	Default rules detected the controlled ICMP sequence
Wazuh ICMP export	20 matching SIEM alerts	Ingestion and correlation preserved expected events
Local rule	SID 1000001 loaded after syntax validation	Analyst-authored logic was active and reproducible
Custom alert exports	One matching Suricata and Wazuh event	Custom URI logic traveled end to end
Hash verification	15 artifacts reported OK	Collected files remained unchanged after hashing

Findings, Limitations, and Recommendations

Confirmed Findings

- Suricata reliably observed traffic on enp0s3 and wrote structured EVE JSON.
- Wazuh preserved core packet, flow, signature, HTTP, and agent fields for analyst triage.
- Default and custom signatures produced the expected 20-to-20 and 1-to-1 alert counts.
- The IDS remained passive; endpoint and Apache availability were not disrupted.

What Was Not Proven

- This exercise did not measure enterprise throughput, false-positive rates, encrypted-traffic inspection, packet loss under load, or production retention requirements.
- A signature match is detection evidence, not attribution or proof of successful exploitation.
- The HTTP 404 response confirms the path was absent; the analytic value came from detecting the request attempt.

Prioritized Recommendations

Priority	Recommendation	Rationale
High	Monitor sensor, interface, EVE output, agent ingestion, and alert freshness	Prevents silent monitoring failure from appearing as clean traffic
High	Version-control local rules and require syntax tests before service restart	Improves reproducibility and reduces deployment risk
Medium	Tune noisy signatures and add thresholds or exception logic	Improves signal quality while preserving coverage
Medium	Build a Suricata dashboard and correlate network alerts with endpoint activity	Speeds triage and strengthens cross-source confidence

Troubleshooting and Engineering Judgment

- Diagnosed a zero-byte EVE log by tracing a service restart loop to the nonexistent eth0 interface; corrected both defaults and AF_PACKET YAML to enp0s3.
- Detected a duplicated Wazuh localfile collector, restored the pre-Suricata backup, reinserted one block, validated exit code 0, and confirmed a single active collector.
- Narrowed HOME_NET from the full /24 to 10.10.10.4/32 so the protected-host boundary matched the lab's analytic intent.

Employer-Facing Project Summary

This project demonstrates network detection engineering from deployment through validation: configure a Suricata sensor, define network scope, collect EVE JSON, integrate with Wazuh, test default coverage, author and validate a custom signature, interpret searchable SIEM fields, troubleshoot ingestion failures, and preserve a verifiable evidence package.

Resume Bullet

PORTFOLIO-READY BULLET

Integrated Suricata IDS with Wazuh EVE JSON collection; validated 20 default ICMP alerts and one custom HTTP-path detection end to end, while preserving 15 SHA-256-verified evidence artifacts.

Interview Talking Points

- Why HOME_NET was narrowed to the single protected server instead of the full lab subnet.
- How an interface mismatch caused a healthy-looking install to produce no EVE telemetry, and how service logs isolated the fault.
- Why the custom HTTP rule fired on a 404 response and how that differs from proving successful access.
- How Wazuh rule level and Suricata severity serve related but distinct prioritization purposes.

Skills Demonstrated

Suricata IDS | EVE JSON | AF_PACKET | Wazuh log collection | Network telemetry | Signature authoring | Detection validation | SIEM triage | Linux troubleshooting | Evidence integrity

AI and Source Transparency

The analyst personally configured the virtual machines, executed the tests, validated the alerts, interpreted the results, captured the screenshots, and preserved the evidence. Generative AI assisted with command sequencing, troubleshooting suggestions, and documentation structure. Technical claims were checked against direct lab evidence and the official references below.

References

- Wazuh — Network IDS integration: <https://documentation.wazuh.com/current/proof-of-concept-guide/integrate-network-ids-suricata.html>
- Wazuh — Localfile configuration: <https://documentation.wazuh.com/current/user-manual/reference/ossec-conf/localfile.html>
- Suricata — Rule format: <https://docs.suricata.io/en/latest/rules/intro.html>
- Suricata — Rule management: <https://docs.suricata.io/en/latest/rule-management/suricata-update.html>
- Suricata — EVE JSON format: <https://docs.suricata.io/en/latest/output/eve/eve-json-format.html>

Official documentation accessed August 28, 2026.